



NumPy Cheatsheet

Import

```
import numpy as np
```

NumPy Arrays

```
menu_prices = np.array([14.99, 5.99, 21.00])
menu_quantity = np.array([5, 8, 1])
```

Indexing

```
print(menu_prices[0]) # Output: 14.99
print(menu_prices[1]) # Output: 5.99
print(menu_prices[2]) # Output: 21.00

print(menu_prices[-1]) # Output: 21.00
```

Slicing

```
boba = np.array([3, 2, 1, 0, 1])

print(boba[0:2]) # Output: [3 2]
print(boba[2:]) # Output: [1 0 1]
print(boba[-2:]) # Output: [0 1]
```

2D Arrays

```
seating_plan = np.array([
    [1, 2, 3],
    [4, 5, 6],
    [7, 8, 9]
])

print(seating_plan[0]) # Output: [1, 2, 3]
```

Array-Array Operations

```
gas = np.array([20.32, 44.87, 67.03])
electric = np.array([80.10, 77.79, 67.73])

total = rent + gas # Adding the two
```

Arithmetic Operations

```
order = np.array([12.99, 9.99, 4.99])

print(order + 2.00) # Addition
print(order - 1.50) # Subtraction
print(order * 0.8) # Multiplication
print(order / 3) # Division
print(order % 2) # Modulo
```

NumPy Functions

```
water = np.array([[6, 2, 3, 8, 7, 9, 7,],
                  [5, 5, 10, 4, 7, 5, 4]])

print(np.min(water)) # Output: 2
print(np.max(water)) # Output: 10
print(np.sum(water)) # Output: 82
print(np.average(water)) # Output: 5.8571
```

Axis

```
np.sum(water, axis=0) # ↓ Down each column
np.min(water, axis=1) # → Across each row
```

.shape

```
print(water.shape) # Output: (2, 7)
```

.reshape()

```
arr = np.array([1, 2, 3, 4, 5, 6, 7, 8])
new_arr = arr.reshape(2, 4)

# [[1 2 3 4]
#  [5 6 7 8]]
```

.arange()

```
arr = np.arange(start=1, stop=10, step=3)
# [1 4 7]
```